





Process Planning and CAPP

seen a lot of different
processes → Goal to make
the final part ⇒ PROCESS
PLANNING = set of activity to define for
going from raw material to final object

Groover M.P.: Chapter 36



The student knows:

- What process planning is;
- What possible approaches to automation are.





PROCESS PLANNING

from Raw material to FINAL OBJECT → if complex object --
How to put together on final object... ← You need to analyze the SUB-Assembly and HOW to put all together!

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Process planning is usually the principal function within the Manufacturing Engineering Department of a company.

It refers to planning the manufacturing processes for economical production of high quality products:

- Deciding appropriate **processes** and their **sequence**;
 - Determining **tooling** requirements;
 - Selecting **equipment**;
 - Estimating **costs**.
- (Detailed description)
- cutting tool + Fixturing + dies for CASTING
- all done considering the COSTS

define how to fabricate each component and how to put all together!
⇒ decide all the STEPS to create the product

We require a given throughput depending on the market

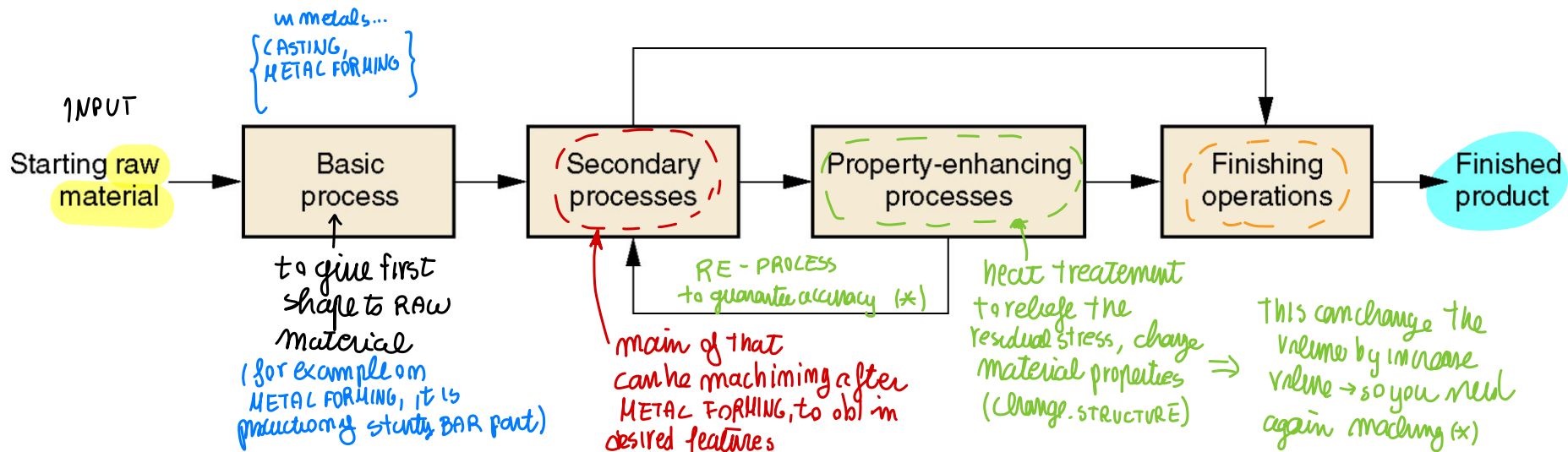


TYPICAL PROCESSING SEQUENCE

schema of overall process planning sequence...

A typical processing sequence to fabricate a discrete part consists of:

1. A basic process to establish initial geometry of part;
2. One or more secondary processes to transform starting geometry into final part shape;
3. Operations to enhance physical properties;
4. Finishing operations.





DETAILS IN PROCESS PLANNING

- **Processes and sequence** - process plan should briefly describe all processing steps used on the work unit in the order they are performed. *select them for each step of process the equipment*
- **Select equipment** to be used - try to develop process plans that utilize existing plant equipment, otherwise, part must be purchased, or new equipment must be installed in the plant. *and specify whatever we need*
- **Specify** what tools, dies, molds, fixtures, and gages are needed. *DESCRIBE all possible details*



DETAILS IN PROCESS PLANNING

define cutting tools and condition (parameters)

- **Specify methods** - hand and body motions, workplace layout, small tools, hoists for lifting heavy parts:
⇒ understand How human moves in the process!
 - Methods must be **specified for manual operations** and manual portions of machine cycles. *automation helps but mainly human help processes*
- **Estimate production costs** - often accomplished by working with cost estimators.
- Specify **cutting tools and cutting conditions** (speeds and feeds) for machining operations.



THE ROUTE SHEET

OUTPUT of process planning

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The document that specifies the details of the process plan:

- The route sheet is to the process planner what the engineering drawing is to the product designer;
- Route sheet should include all manufacturing operations to be performed on the workpart, listed in the order in which they are to be performed.

Part No: 031393		Part Name: Housing, valve		Rev. 2	Page <u>1</u> of <u>2</u>	
Matl: 416 Stainless		Size: 2.0 dia × 5. long		Planner: MPG		Date: 3/13/XX
No.	Operation	Dept.	Machine	Tooling, gages	Setup time	Cycle time
10	Face; rough & finish turn to 1.473 ± 0.003 dia. × 1.250 ± 0.003 length; face shoulder to 0.313 ± 0.002; finish turn to 1.875 ± 0.002 dia.; form 3 grooves at 0.125 width × 0.063 deep.	L	325	G857	1.0 h	8.22 m
20	Reverse; face to 4.750 ± 0.005 length; finish turn to 1.875 ± 0.002 dia.; drill 1.000 + 0.006, −0.002 dia. axial hole.	L	325		0.5 h	3.10 m
30	Drill & ream 3 radial holes at 0.375 ± 0.002 dia.	D	114	F511	0.3 h	2.50 m
40	Mill 0.500 ± 0.004 wide × 0.375 ± 0.003 deep slot.	M	240	F332	0.3 h	1.75 m
50	Mill 0.750 ± 0.004 wide × 0.375 ± 0.003 deep flat.	M	240	F333	0.3 h	1.60 m

that OUTPUT contains all element
needed during the process (to do the part)
↓ documents describing set of operations
DESCRIBE ALL
(grouped in phases and sub-phases)
↓
Need to know everything!



THE ROUTE SHEET

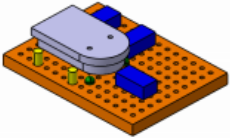
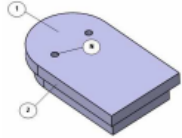
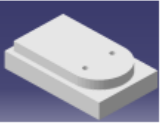
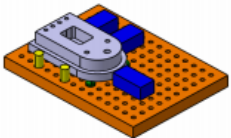
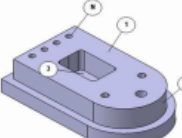
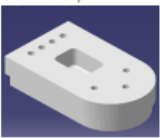
(Example)

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Simple situation
High level Route
Sheet for
machining

Starts from
a block...

Set operation to
perform! →
Detailed ROUTE sheet
for each operation

Phase	Position (Fixturing Assembly)	Machined features	Operation sequences	Tool (mill/drill)
1	- Pictures of the assembly 1 	- Pictures of the part  - Pictures of the resulted part (Saved from the simulation) 	1. Facing 2. Contouring N. drilling	tool 1 tool 2 tool N
...	...	...	...	...
N	- Pictures of the assembly N 	- Pictures of the part  - Pictures of the resulted part (Saved from the simulation) 	1. Facing 2. Contouring 3. pocketing N. drilling	tool 1 tool 2 tool 3 tool N

How to position,
describe FIXTURING...

positioning inside
machine TOOL you
have dedicated fixturing
for the part

↓
with some clamping
keeping properly the part

Set of Decisions

No.	Features	Phase	Tool type	Tool data					a_p	a_e	V_c	n	V_f	K_c	P_c	OK/NO
				D	l1	max a_p	max a_e	...								
1	Facing	1	face mill	5	1	1	1	...	0.5	1	20	100	2	1500	100	OK
...																
No.																



AN EXAMPLE

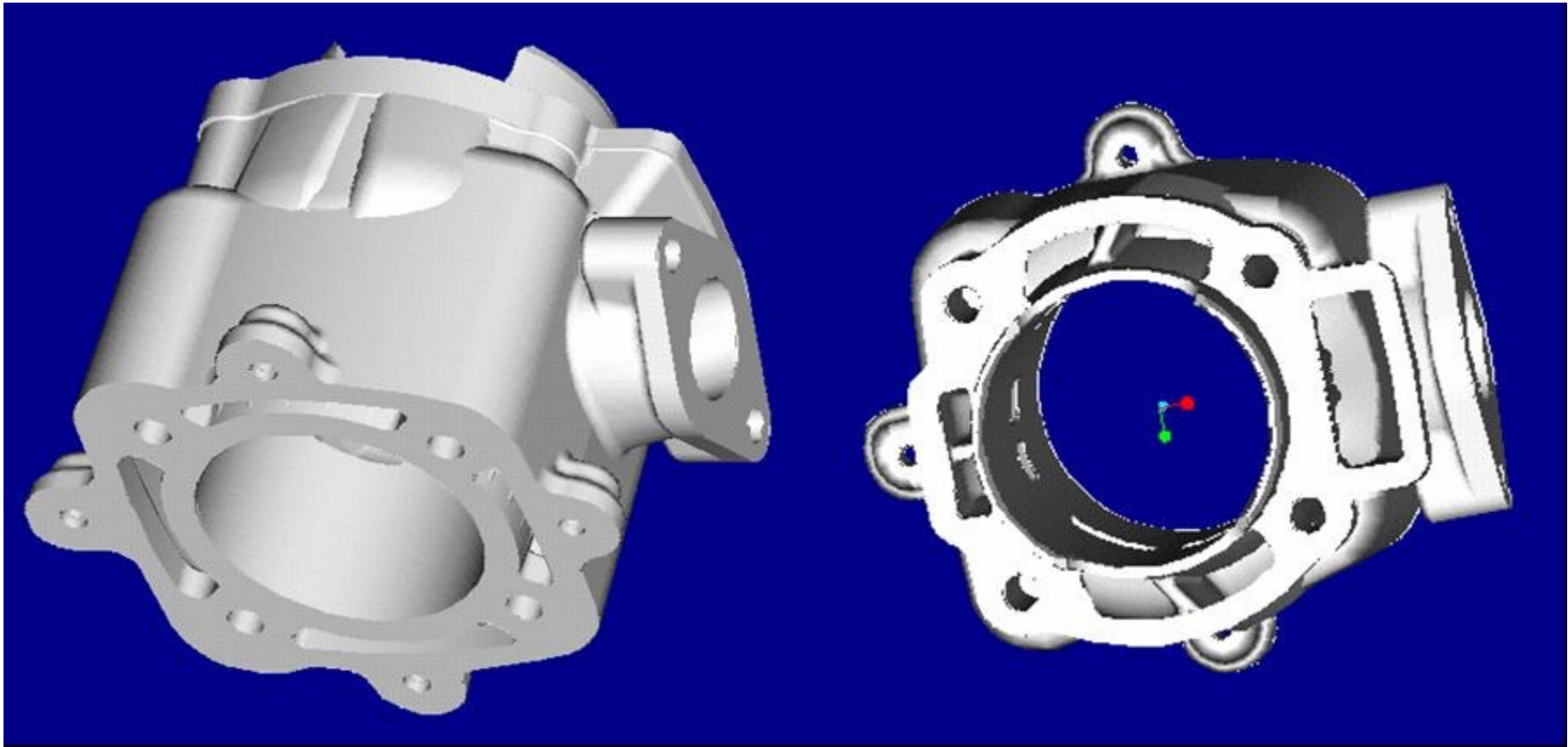
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complex part :

final obj coming from casting
(engine box of motorcycle)

DM PROCESS-PLANNING
you plan CASTING,
MACHINING...

↳ TAKE Decision...



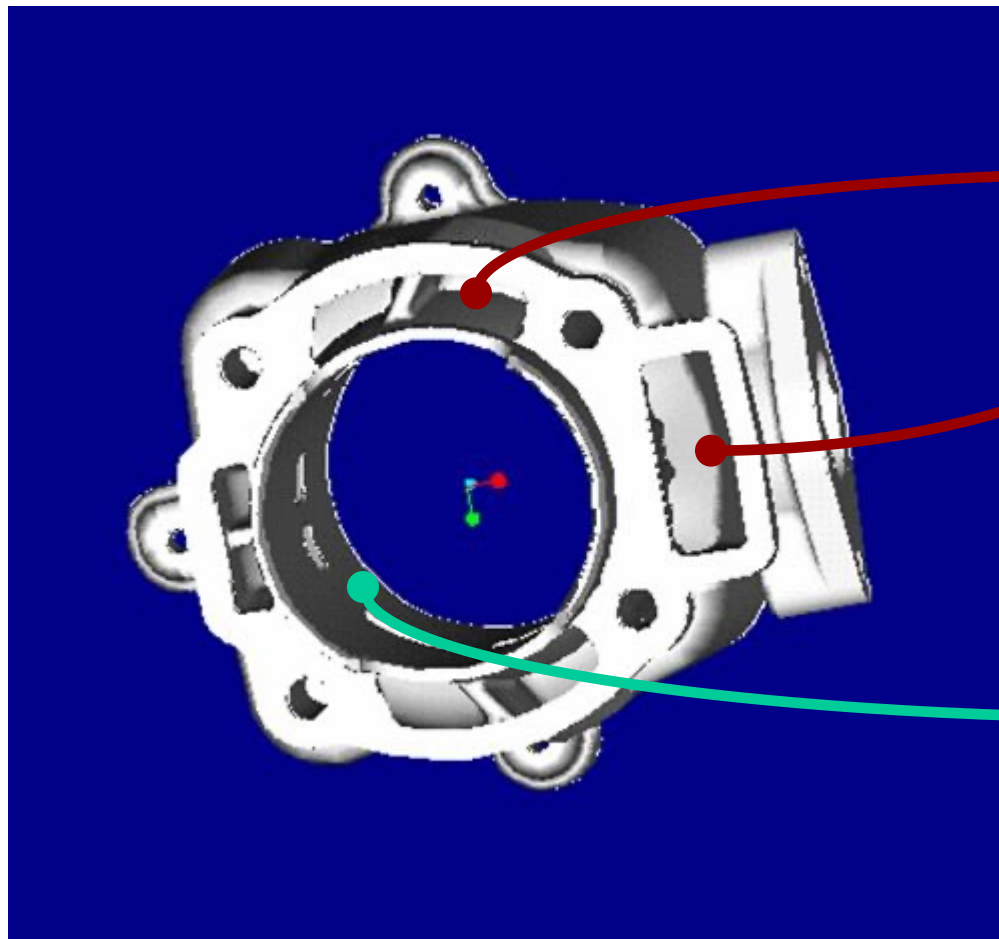


AN EXAMPLE

Take Decision → what and how to machine

10

↓ some features can remain
ROUGH, BUT internal
cylinder is Relevant, it
need finishing



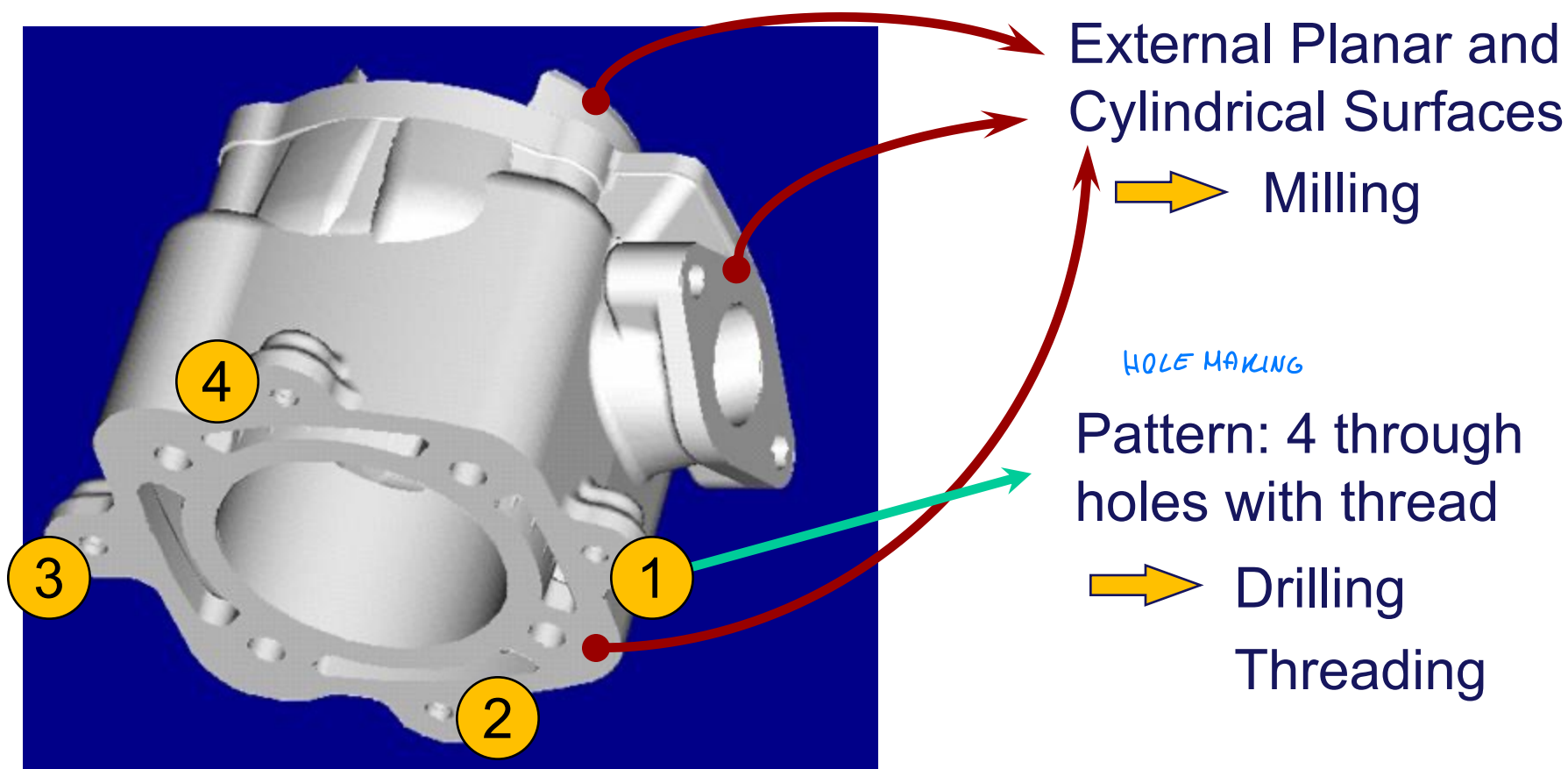
Raw Through Pocket

→ No Operation

+ HOLES to do on the part

Through Hole

→ Finishing





AN EXAMPLE

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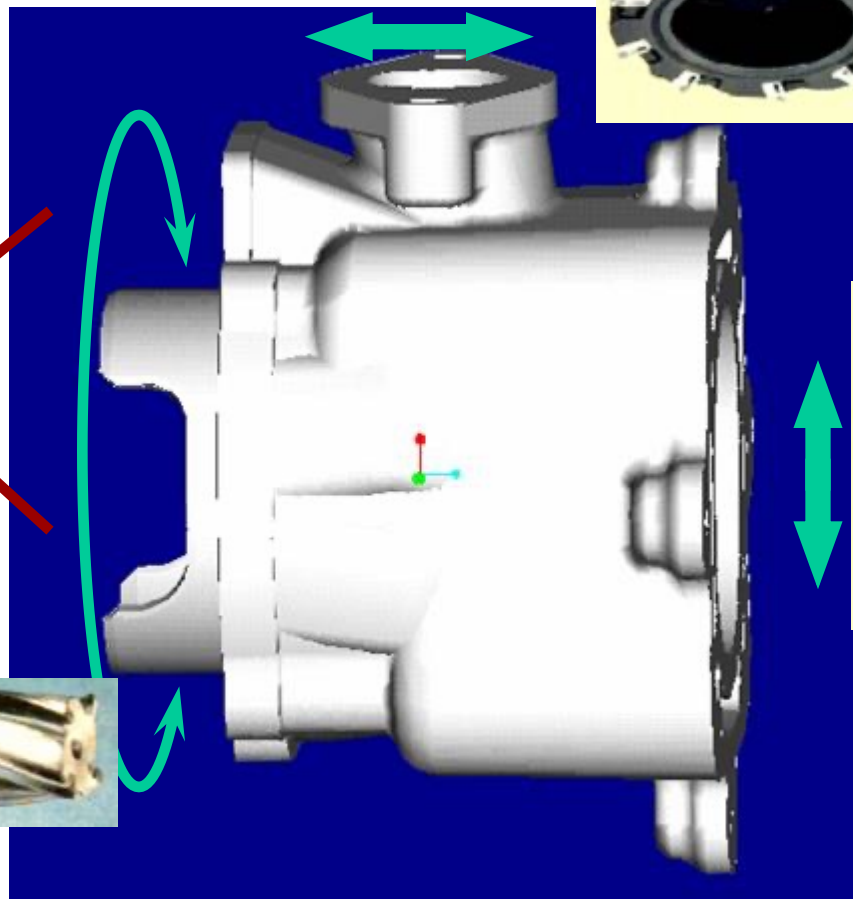
Milling

external plane requiring MILLING
(creating plane + external cylindrical
surface (CONTOURING of surface))

Understand which tools to

use, tool
requirement +
machining time

avoid changing
of the TOOL, plan
well the sequence
(common tool for
more
features)



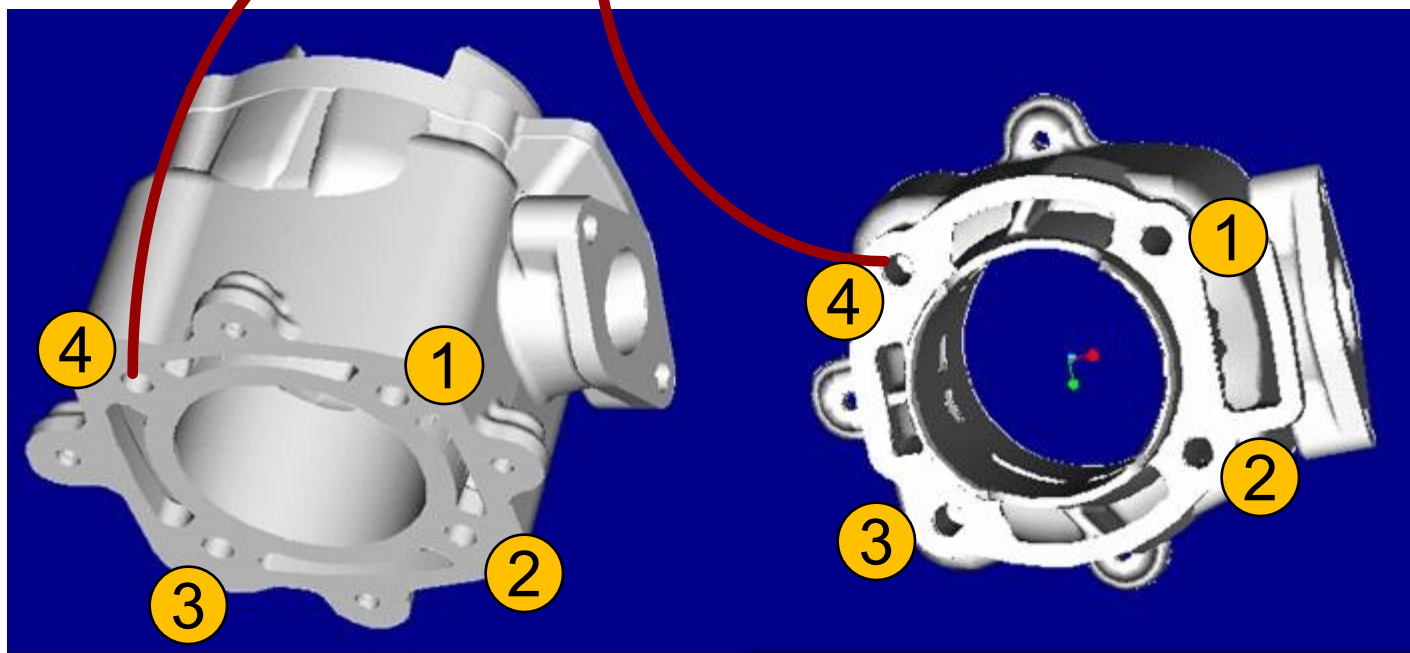


4 Through Hole Pattern



Drilling

this can be done in two ways!

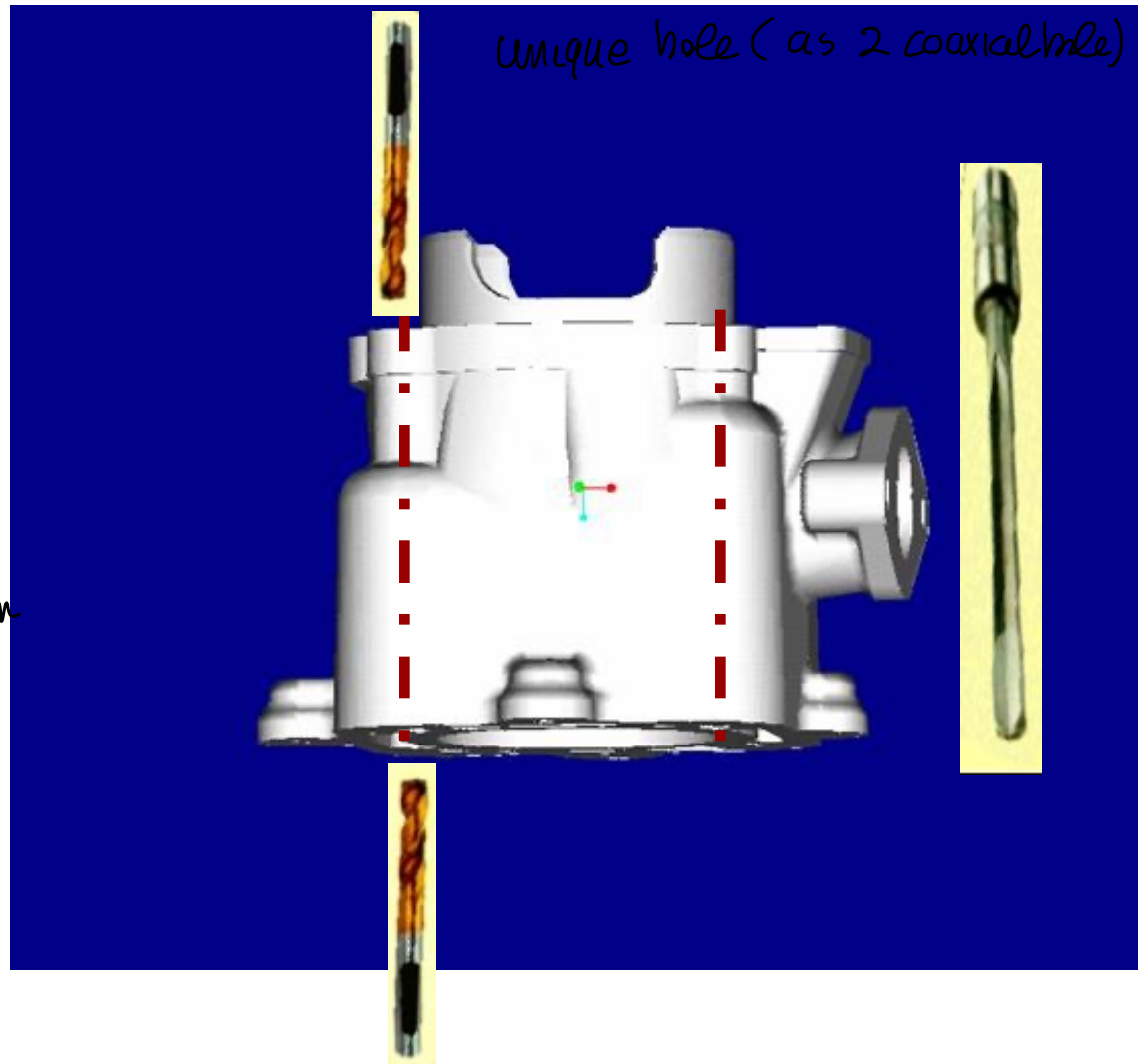




Drilling

Standard
Drilling

*you can obtain
the hole in
2 steps*



unique hole (as 2 coaxial hole)

Deep
Hole
Drilling

*or obtain
the hole
all in one step*



INSPECTION PLANNING

(as on CHU)
you have Requirements to verify on the part... so some Capability

Part Nominal Geometry and Geometrical Product Specifications

[measure the feature]
Coordinate Measuring System (CMS)

Process Capability



Inspection Planning

What to inspect

When to inspect

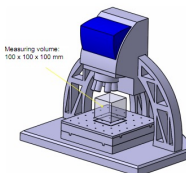
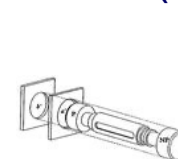
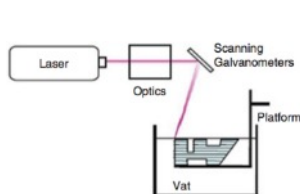
How to inspect

it can be done along the FULL Process
↓
it is better to look quality more times to CORRECT

What kind of device, which PROBING? how to sample the part..



Good Systems Capability
↓
understand what to inspect → when perform INSPECTION





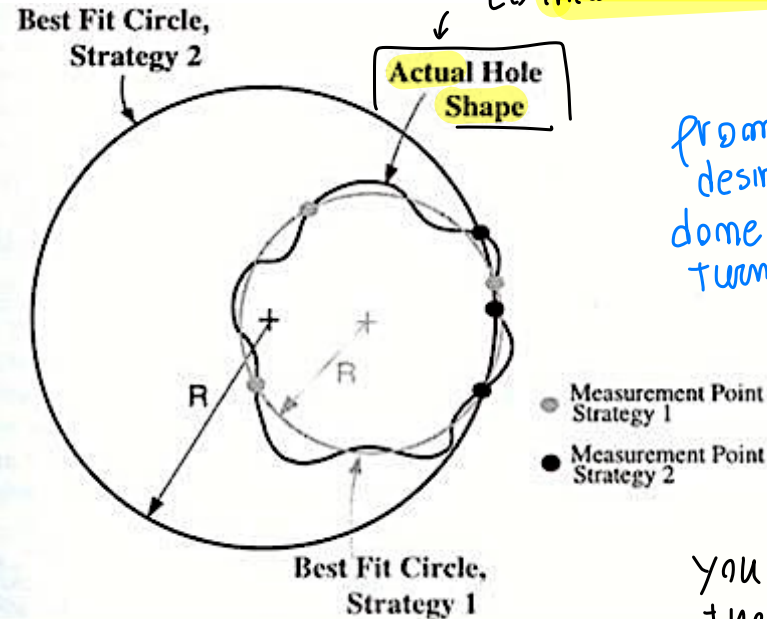
SAMPLING STRATEGY

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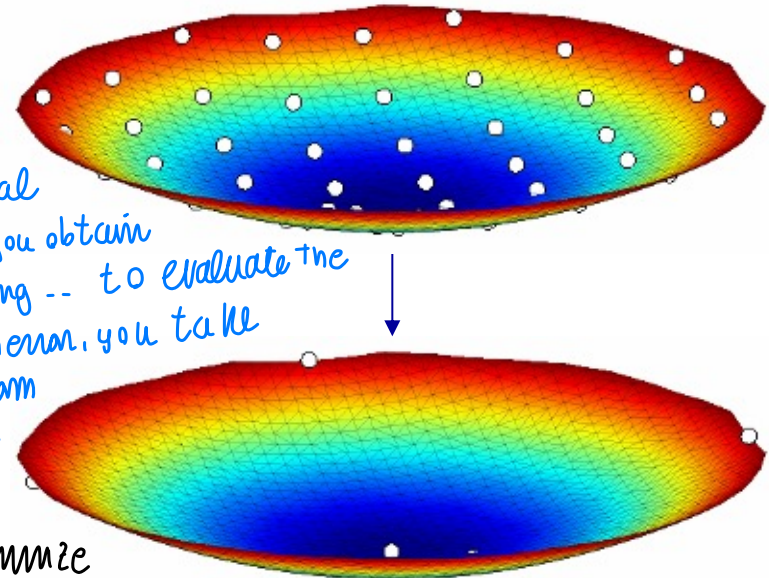
} sampling the surface
(like decreasing machining parameters)}

⇒ if you make a sampling error ... to have a circle you need 3 points -- and if they are close to each other, you obtain a wrong fitting
(NOT ROBUST ALGORITHM)

to make an HOLE



from the desired plane done by radial turning → you obtain bending -- to evaluate the deformation, you take Random points



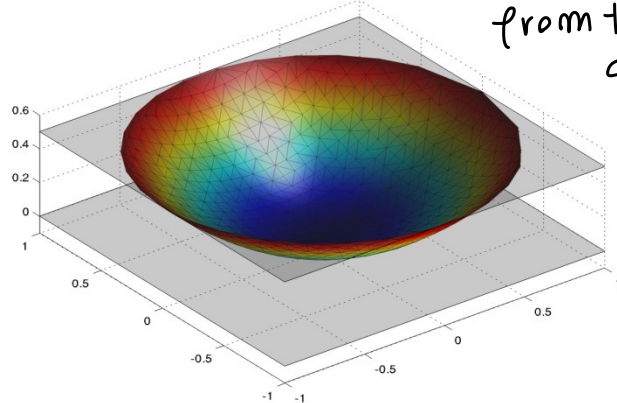
you can minimize the taken points if knowing what to expect (better info)



SAMPLING STRATEGY

Different sampling strategies may yield different error evaluations.

Real surface

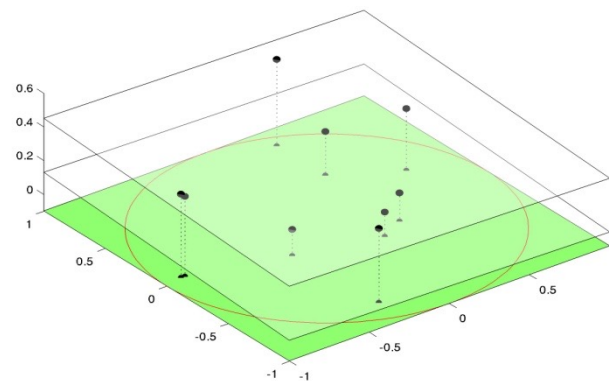


from that particular
concavity $\rightarrow$

$$e_{real} = 0.5$$

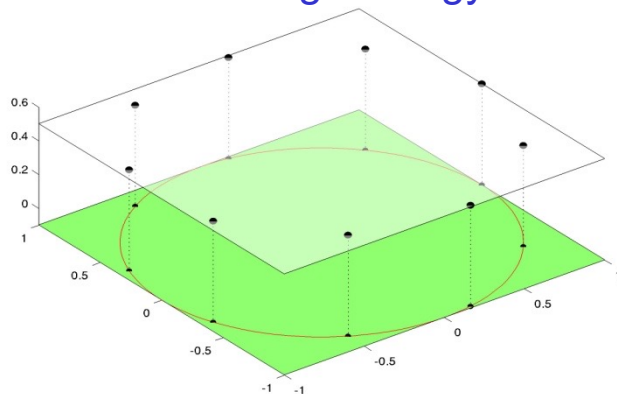
Random strategy

depends on collected points, underestimates
the real value



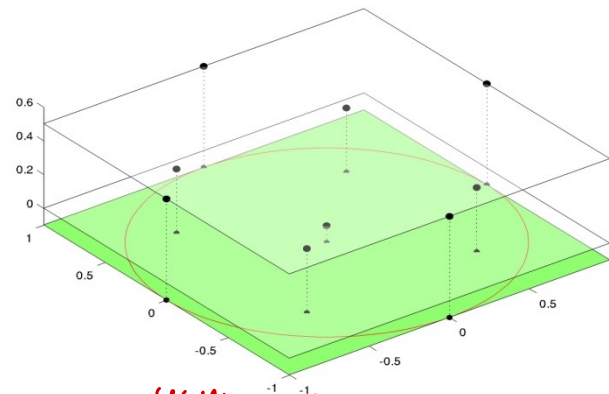
$$\hat{e} = 0.32$$

Wrong strategy



$(\hat{e} = 0)$
TOTALLY
uncorrect

Good strategy



$$\hat{e} = 0.5$$

using an idea of the
probable shape final!



INSPECTION COST FUNCTION

you have cost in INSPECTION

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you may consider costs → use few points to estimate the errors (more points means more cost)

Inspection Errors
Cost C_E

Cost derived from inspection

errors of type A → NOT enough INFORMATION
(indicate as bad a good part)

and type B
(indicate as good a bad part)

↳ classical error

(A) Removing in the process a part that we consider BAD but instead is good → all production cost is loss

(TRADE-OFF)

(B) saying a part is good but it's BAD, if you buy a BAD object that doesn't work → manufacturer must change all, HUGE COST!

Inspection Cost
 C_I

Measurement
Cost C_M

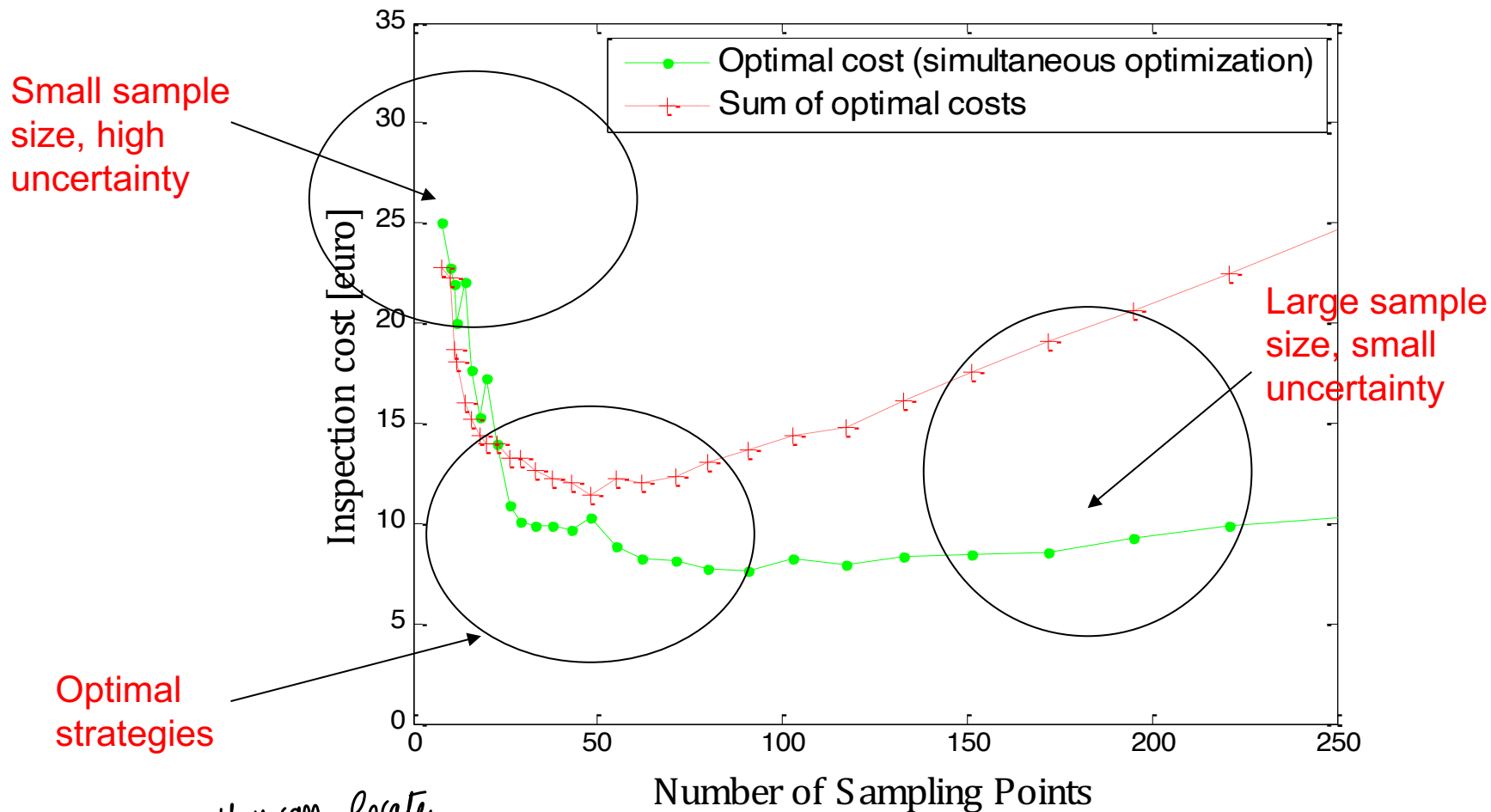
Cost arising directly from performing a measurement
(cost for POINT collection)

Well locating sampling points give you a

TRADE OFF IN SAMPLING?

SAMPLING STRATEGY

- Increasing the SAMPLING POINTS → you INCREASE the INSPECTION COST -- you have a curve like that
after some point... it costs a lot



you can locate points in different STRATEGY $\Rightarrow$ Inspection cost = $C_u + C_e$



PROBE CONFIGURATION

issue in deciding how to sample

+ other decision during MEASUREMENT/VERIFICATION

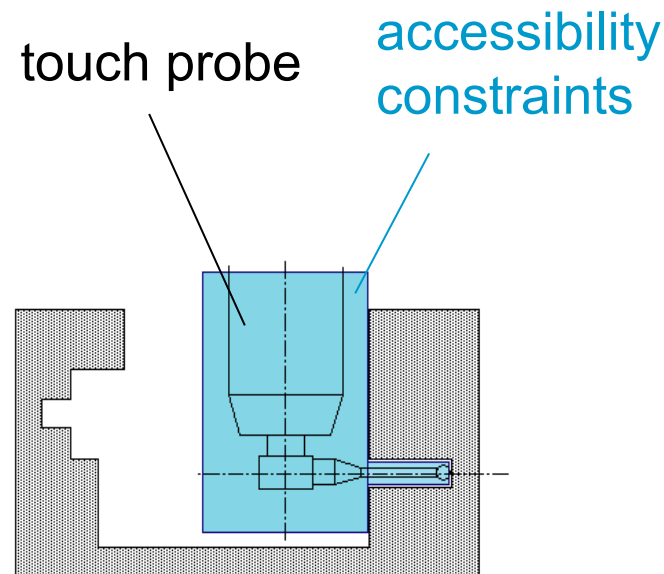
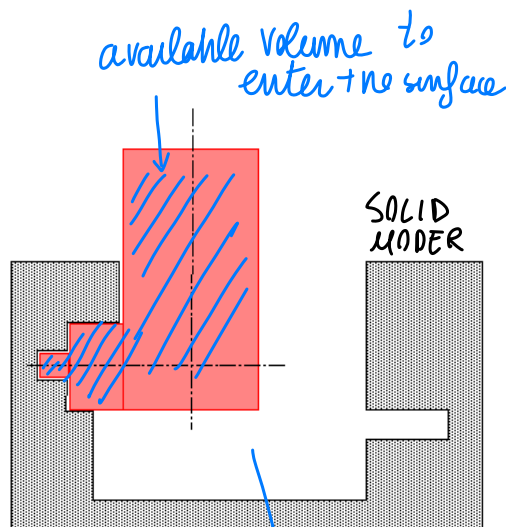
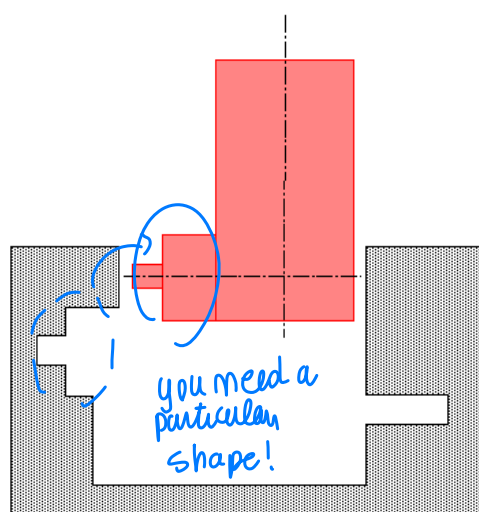
Configure a touch probe means to determine its more suitable **orientation, structure, dimensions** to inspect the part.

VARIABLE PROBE configuration → different solution ⇒ How to access the features of the part?

Two main activities:

- Accessibility Analysis
- Clustering and Configuration

← how to reach the surface (access surface) depending on PROBING system



in X-RAY: how many Img to take? - -

configure a probe to stay inside this volume



PART ORIENTATION AND FIXTURE CONFIGURATION

*you can position the part in different ways
different config. of PROBING head, depending also on the part*

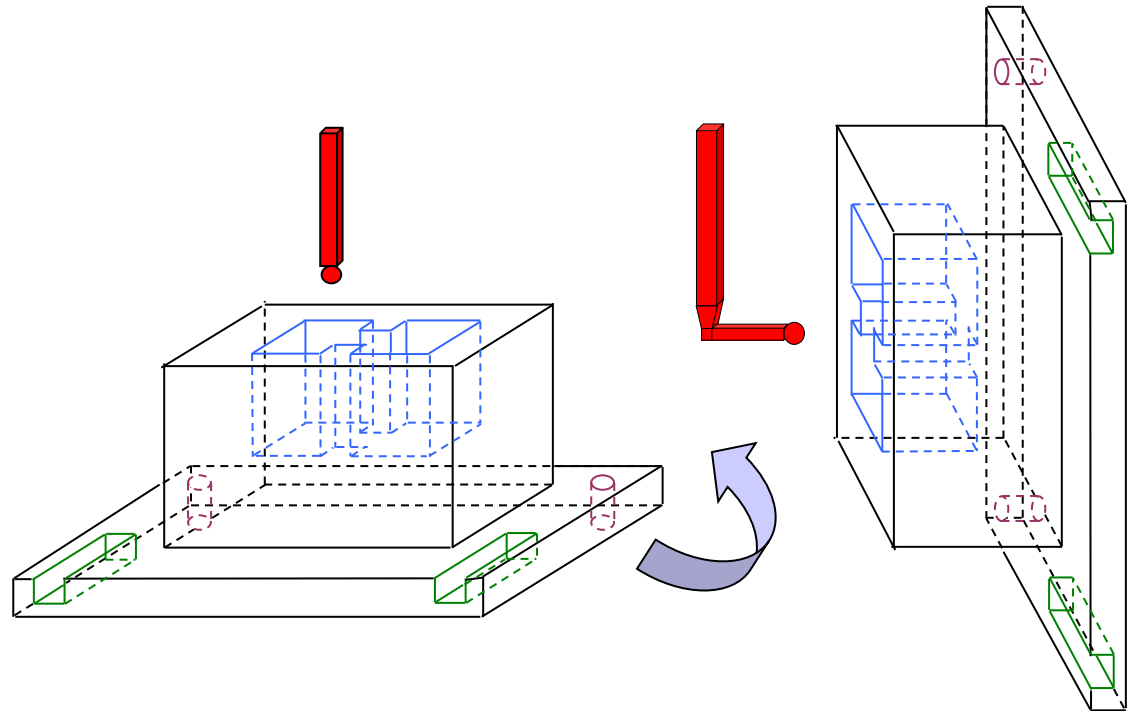
Configure a touch probe
means to determine its more
suitable

**ORIENTATION,
STRUCTURE,
DIMENSIONS**
to inspect the part.

*Sometimes
you have to
change the PROBE!
configure the TOOL...*



PART ORIENTATION





Decision to make for PROBE choice, modular elements to put together

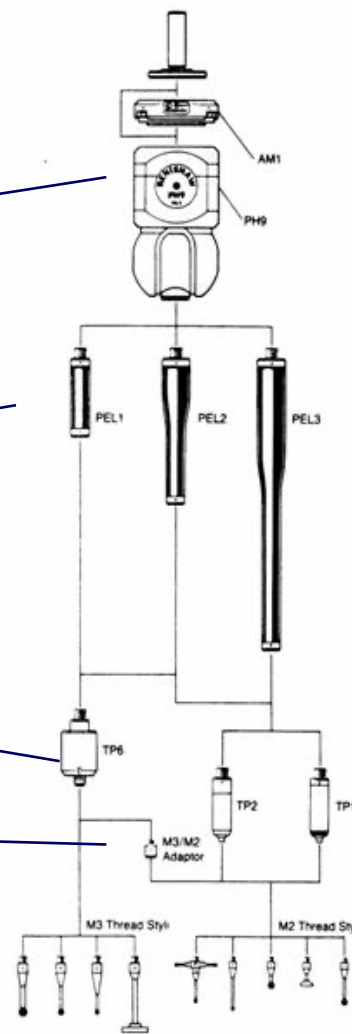
Probe Head

Probe Extension

Probe

Accessory / Stylus Extension

Stylus



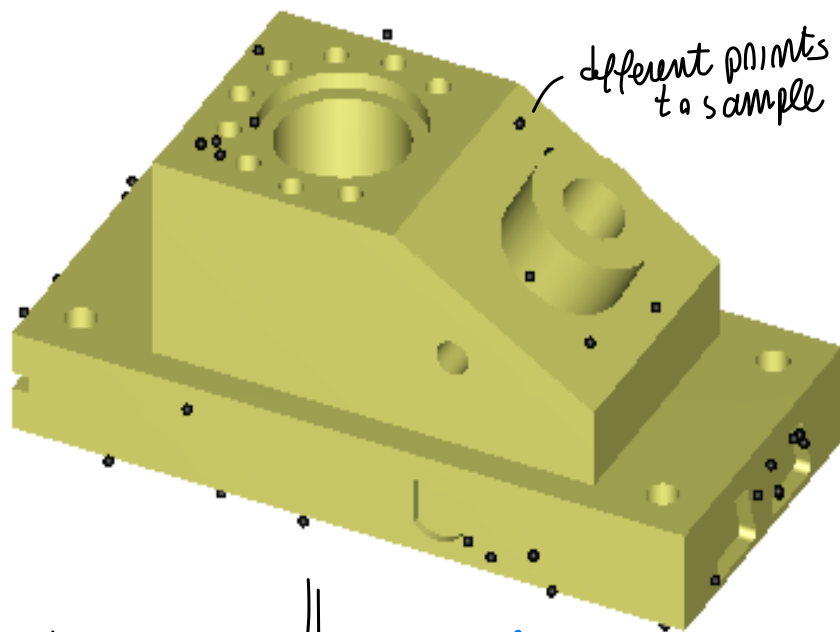
Big TIP = STIFF stylus BUT physical filter during measurements



CMM PATH PLANNING

POINT to POINT sampling
decide & feature, How to sample those surfaces

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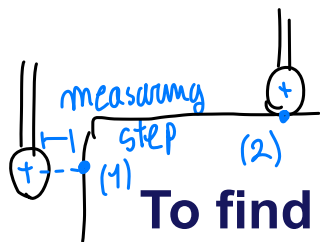


Traveling Salesman Problem

Graph Nodes: measuring points

Arcs: collision free trajectories between each couple of points.

There is not a priori knowledge of the arc length!



PATH planning to sample points (1), (2) ⇒

N points to sample! We need in CMM path planning → from cloud of points to reach we need to define the sequence of measuring the points!

To find an optimal solution to a TSP problem is NP-hard.

The complexity of an exhaustive search is $N!$

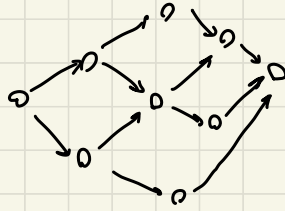
A priori knowledge: $N(N-1)/2$ collision free trajectories.

e.g. $N = 50$, 1225 trajectories

TSP PROBLEM

ISSUE you don't know how to go from one point to another

TPS PROBLEM $\Rightarrow$ to handle this PROBLEM you have
a decisional GRAPH



NP-HARD
PROBLEM!

Missing knowledge
a PRIORI $\rightarrow$ you don't know
the TIME COST to move
 $\swarrow$ between points!

usually
NO CONSTRAINTS! to move between

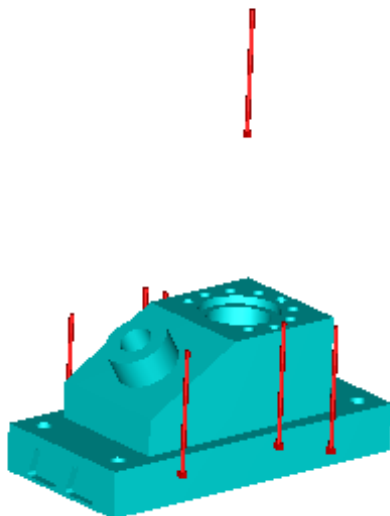
FIRST compute collision trajectory to evaluate the cost

$N=50 \rightarrow 1225$ traj Comedumy POINTS $\rightarrow$ full GRAPH

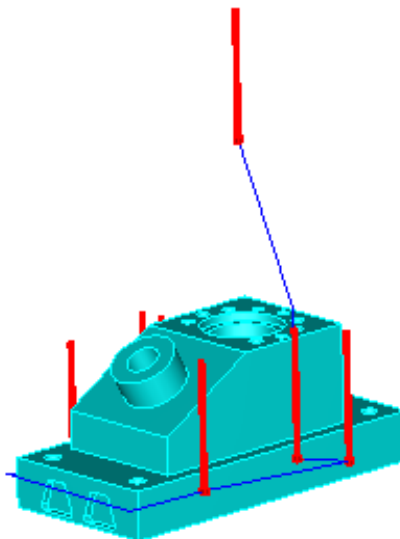
{ complex
OPTIMIZATION! } $\rightarrow$ how to move
machine, PATH
Planning...

Representation
from which solve the problem!

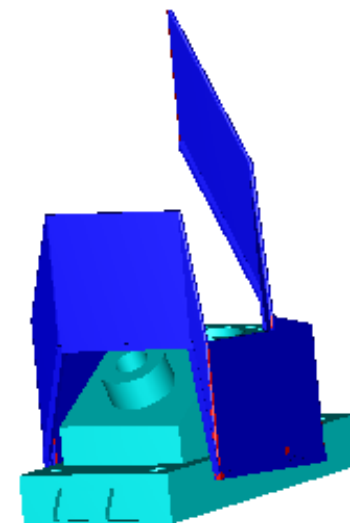
WE need to solve this complex combinatorial problem



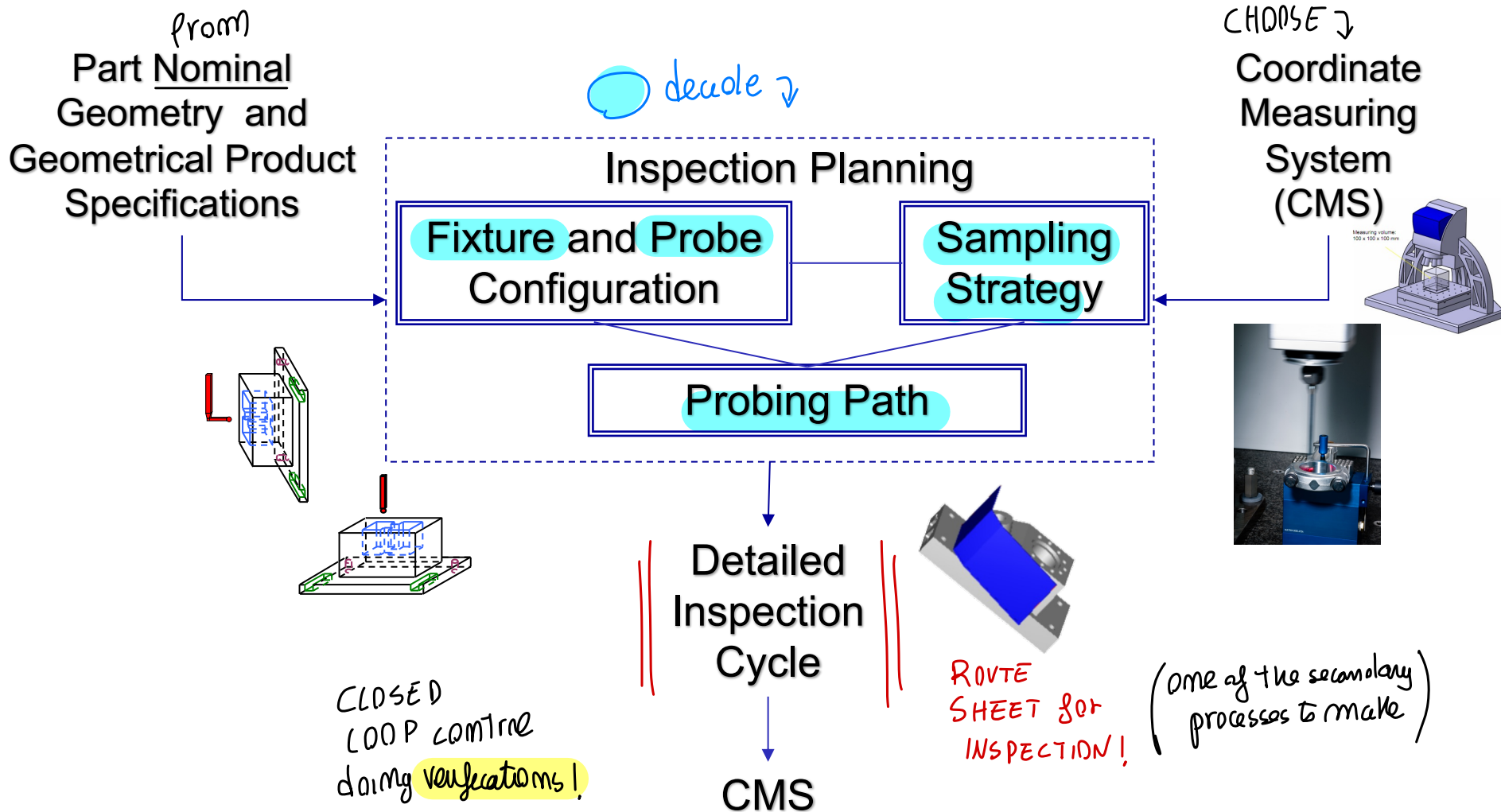
Measuring points on
ANSI/ANC101



Tip path planning
Collision free trajectory



Probe path planning.



KNOWLEDGE-BASED MANUFACTURING (ISSUE)

↓ during PLANNING → main PROBLEMS inside manufacture environ

- Shop people knowledgeable in manufacturing processes are gradually retiring. → increase on experience BUT than RETIRING
- Different manufacturing people create different solutions (process plans) to solve the same problem (divergence problem). well experience people provide different solution! DIFFERENT PLANNING! (XX ?)
- What happens if a new product or a new technology is considered? no experience in new tech! NOT experience of EXPERT in new TECH of manufacturing

⇒ extract knowledge FROM people → to do OPTIMIZATION during process planning → extract OBJECTIVE FUNCT

An alternative approach to use knowledge is needed:

- During the last several decades, there has been considerable interest in automating the process planning function using computer systems;
- Computer-aided process planning (CAPP) systems provide this alternative.

↙ obtain sw for take decision



Computer-aided process planning systems are designed around either of two approaches:

- 1. Variant systems;
- 2. Generative systems.

Good consistent approach

extract
knowledge → B.B. Mach. Learning solution

? NOT always good, sometimes you should
GO INSIDE the problem!

physical situation → Model

||
{ difficult usage of
JUST M.L. N.N to
plan the process! } without using the KNOWLEDGE... xx ~~~

KNOWLEDGE
BASED
MANIF. !

better than
computer based



(RETRIVAL)

VARIANT CAPP SYSTEMS

Classical approach → When having a PROBLEM... understand similar situation to 28
→ solve it, using PAST knowledge → approaching new PROB

use experience to apply yet known solution

A variant CAPP system is based on an approach called **group technology (GT)**: (well defined methodology)

- Recognizing that although the parts are different, there are families of parts that possess similarities (similar problem = similar solution)
 - When these similarities are exploited in design and production, overall efficiency is improved.

It is also known as *retrieval CAPP systems*. A standard process plan is stored in computer files for each part code number:

- The standard plans are based on current part routings used in the factory, or on an ideal plan prepared for each family;
- For each new part, the standard plan is edited if modifications are needed.



GROUP TECHNOLOGY (GT)

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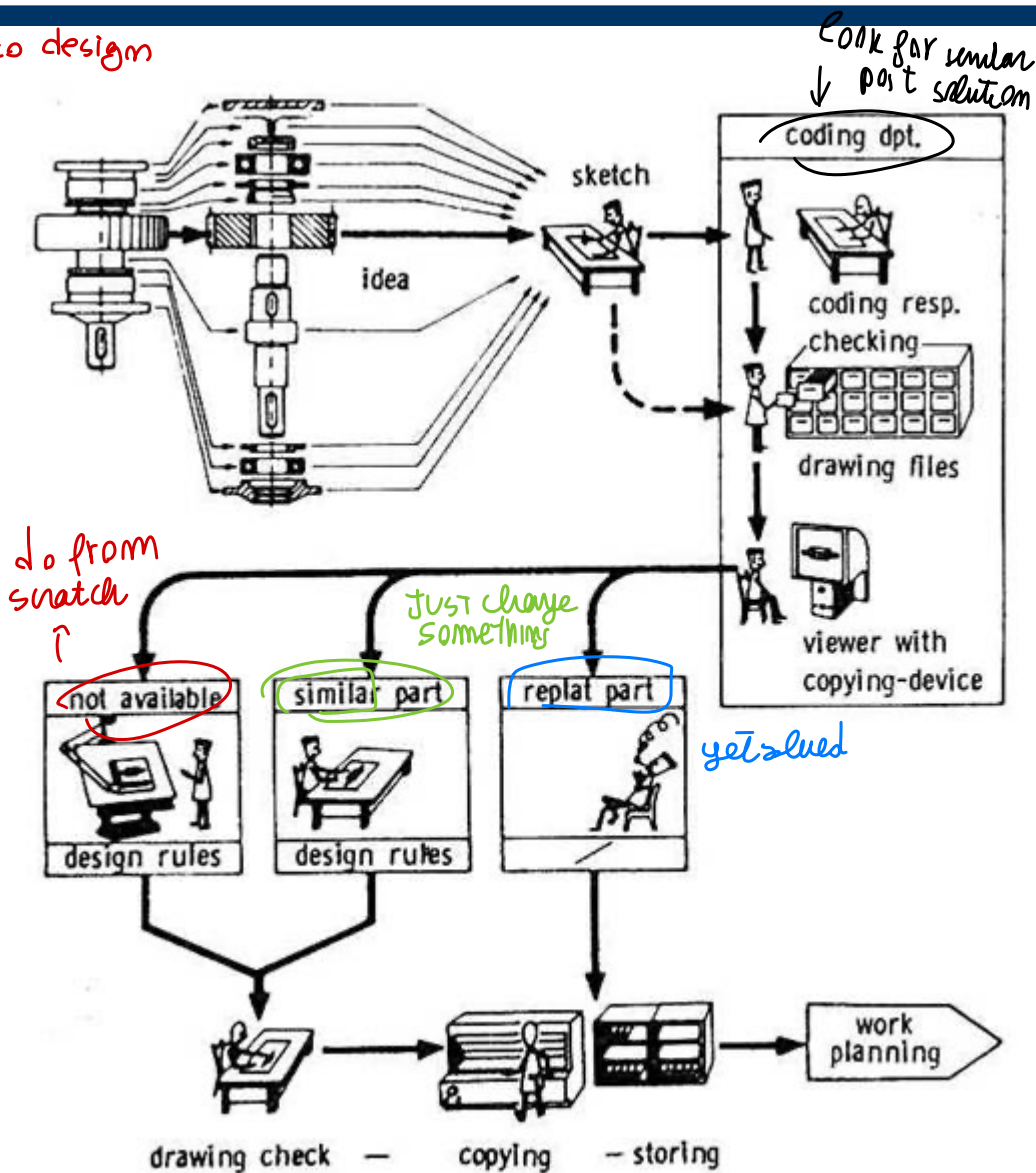
IF you have a PART to design
knowing functionality

Conceptual Design

FIND IF THERE IS YET
AN EXISTING SIMILAR
SOLUTION

Similar Product
Retrieval

Design Variation





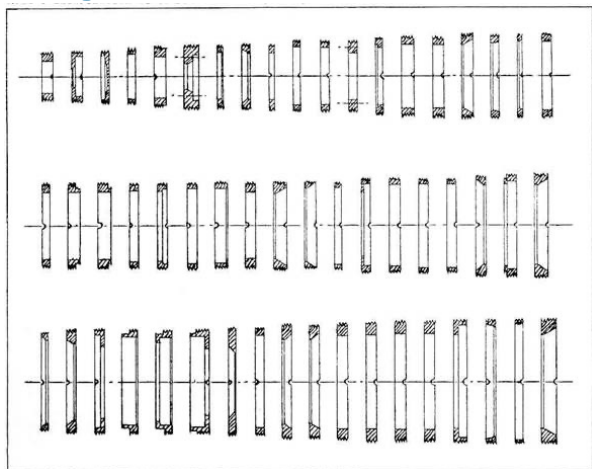
VISUAL BASED Similarities

A group of parts that possess similarities in shape and size, or in the processing steps used in their manufacture:

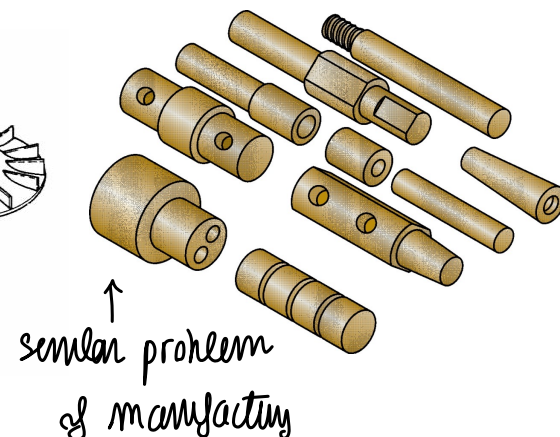
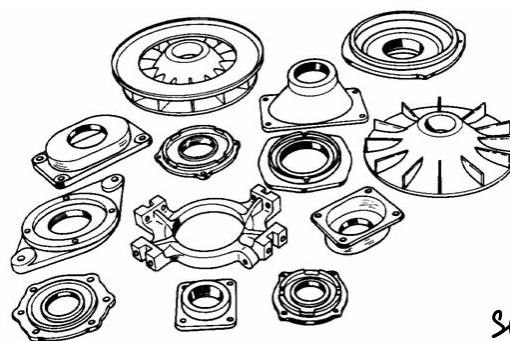
- There are always differences among parts in a family, but the similarities are close enough that the parts can be grouped into the same family.

SIMILAR PARTS = similar machining operations → GROUP the PART in Families

Design Part Families



Manufacturing Part Families





WAYS TO IDENTIFY PART FAMILIES

TO GROUP THE parts..

- (1) Visual inspection *↘ CLUSTER parts looking at it*
 - Using best judgment to group parts into families, based on the parts or photos of the parts.
- (2) Production flow analysis *↘ CLUSTERING algorithm in MANUFACTURING systems*
 - Using information contained on route sheets to classify parts into manufacturing families.
- (3) Parts classification and coding
 - Identifying similarities and differences among parts and relating them by means of a coding scheme.
define a way to classify part based on CODING system
↑
code so similar parts have similar codes



PRODUCTION FLOW ANALYSIS

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Machine	Pièces N°																																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	
K	✓		✓																	✓			✓		✓											
b		✓					✓			✓		✓	✓					✓						✓			✓				✓					
c	✓		✓		✓									✓		✓													✓		✓					
d		✓					✓					✓	✓											✓			✓								✓	
e								✓						✓		✓																			✓	
f								✓						✓		✓			✓			✓													✓	
g	✓		✓		✓										✓	✓				✓		✓					✓								✓	
h	✓	✓		✓													✓		✓		✓		✓													
i								✓						✓				✓			✓		✓				✓									
j								✓						✓		✓		✓		✓		✓				✓										
k				✓		✓			✓		✓																									
l				✓		✓			✓		✓																									
m		✓												✓	✓																					
n		✓					✓			✓		✓	✓																						✓	
o				✓		✓				✓		✓																								
p				✓		✓			✓		✓																									
q	✓		✓		✓											✓		✓																	✓	
r		✓								✓	✓			✓	✓																				✓	
s				✓		✓				✓		✓																								
t							✓							✓																						

Machine															
	2	12	13	24	27	31	7	10	18	1	3	5	15		
b	✓	✓	✓	✓	✓	✓	✓	✓	✓						
d	✓	✓	✓	✓	✓	✓	✓	✓	✓						
m ^o	✓	✓	✓	✓	✓	✓	✓	✓	✓						
n	✓	✓	✓	✓	✓	✓	✓	✓	✓						
R	✓	✓	✓	✓	✓	✓	✓	✓	✓						
2															
c															

2

↓ CLUSTER the part by INSPECTING the MATRIX

respect the machine they go through

Machine	Pièces N°																																		
	2	12	13	24	27	31	7	10	18	1	3	5	15	17	20	23	25	29	4	6	9	11	21	28	35	30	32	33	8	14	19	22	26	16	34
b	✓	✓	✓	✓	✓	✓	✓	✓	✓																										
d	✓	✓	✓	✓	✓	✓	✓	✓	✓																										
m ⁰	✓	✓	✓	✓																															
n	✓	✓	✓	✓	✓	✓	✓	✓	✓																										
R	✓	✓	✓	✓		✓	✓	✓	✓																										
a																																			
c																																			
g																																			
h																																			
Q																																			
K																																			
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P																																			
S																																			
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e																																			
f																																			
i																																			
J																																			

same in planning operations, CLUSTER the OPERATIONS

CLUSTERING tells you the similar machine used, to well organize the LAYOUT of the machines

cells dedicated for a part production → using same machine → GROUP together the machine EASIER managing



Most classification and coding systems are one of the following:

- Systems based on part design attributes;
- Systems based on part manufacturing attributes;
- Systems based on both design and manufacturing attributes.

Design attributes

Major dimensions
Basic external shape
Basic internal shape
Length/diameter ratio
Material type
Part function
Tolerances
Surface finish

Manufacturing attributes

Major process
Operation sequence
Batch size
Annual production
Machine tools
Cutting tools
Material type



PARTS CLASSIFICATION AND CODING

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(Done by each company)

Classification and coding system according to Opitz.

own classification system for each company

many digit for shape + dimension + accuracy --

MILLING machining

Form Code						Supplementary Code				
1st digit Part class		2nd digit Main shape	3rd digit Rotational surface machining	4th digit Plane surface machining	5th digit Auxiliary holes, gear teeth, and forming	Digit				
						1	2	3	4	
0	Rotational parts	$\frac{L}{D} < 0.5$	External shape, external shape elements	Internal shape, internal shape elements	Plane surface machining	Dimension	Material	Original shape of raw material	Accuracy	
1		$0.5 < \frac{L}{D} < 3$								Auxiliary holes
2		$\frac{L}{D} > 3$								
3		$\frac{L}{D} < 2$ with deviation	Main shape	Rotational machining, internal and external shape elements	Auxiliary holes, gear teeth, and forming					
4		$\frac{L}{D} \leq 2$ with deviation								
5		Special								
6	Nonrotational parts	$\frac{A}{B} < 3, \frac{A}{C} > 4$ Flat parts	Main shape	Principal bores	Plane surface machining	Auxiliary holes, gear teeth, and forming				
7		$\frac{A}{B} > 3$ Long parts	Main shape							
8		$\frac{A}{B} < 3, \frac{A}{C} > 4$ Cubic parts	Main shape							
9			Special							



COMPOSITE PART CONCEPT

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once given the CODE
→ ASSOCIATE the part FAMILY to
the PROCESS PLAN

A composite part for a given family is a hypothetical part that includes all of the design and manufacturing attributes of the family:

- An individual part in the family will have some of the features of the family, but not all;
- A **process plan for the part family** would consist of operations required to make the composite part:
 - Such a plan would be able to produce any family member, by omitting operations corresponding to features not possessed by that part.

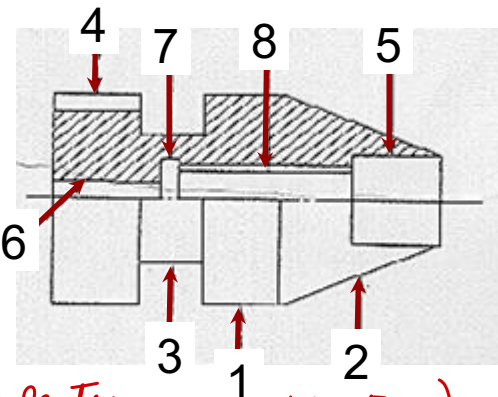
↓ Inherent the Result

define: **Composite Part**

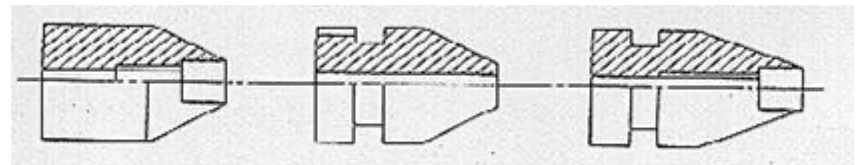
with BASIC machining operations

FAMILY of PARTS

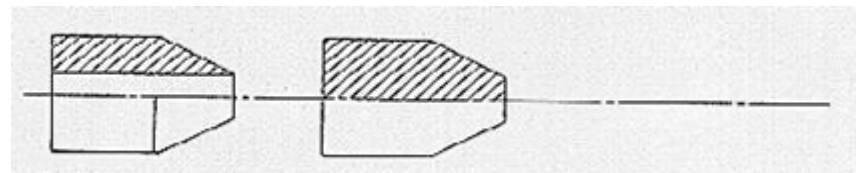
Real Parts (OPTIMAL PROCESS plan & part family!)



all features (+ combinations)



(1,2,5,6,8) (1,2,3,4,5) (1,2,3,5,6,8)



(1,2,5)

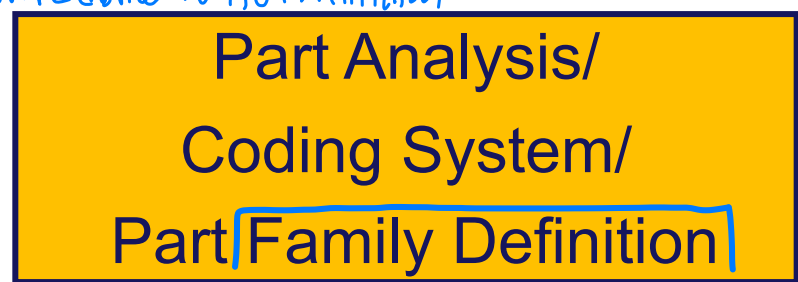
(1,2)



Use Knowledge! VARIANT CAPP SYSTEMS

TO BUILD a system for
decide PROCESS you have 2 stages

IT can be done ~ AUTOMATICALLY



(stage 1)
Preparatory Stage

once you have family knowledge



(Stage 2) Operational Stage



GENERATIVE CAPP SYSTEMS

System uses those rules to take decision

↓ define RULES, formalize knowledge through SET of RULES to take decisions

- Rather than retrieving and editing an existing plan, the process plan is created using systematic procedures that might be applied by a human planner. (not used on AI)
- In a fully generative CAPP system, **the process sequence is planned without human assistance and without predefined standard plans.**
- Designing a generative CAPP system is a problem in

(A.I) → expert systems - computer programs capable of solving complex problems that normally require a human with years of education and experience.

↓ represent problem

Computer → represent knowledge Rule based + Expert syst.

INFERENCE...

Understand Rules and different Rouths of processing, different choices possibility



COMPONENTS OF AN EXPERT SYSTEM

- **Knowledge base** - *We have RULES on the PROBLEM → to formalize* The technical knowledge of manufacturing and logic used by process planners must be captured and coded in a computer program.
- **Computer-compatible part description** - The description must contain all the pertinent data needed to plan the process sequence.
- **Inference engine** - The algorithm that applies the planning logic and process knowledge contained in the knowledge base to a given part description.

more software to
visualize **Rules** → EXPERT
↑
SYSTEMS

for our
environment → adapting to
company knowledge!

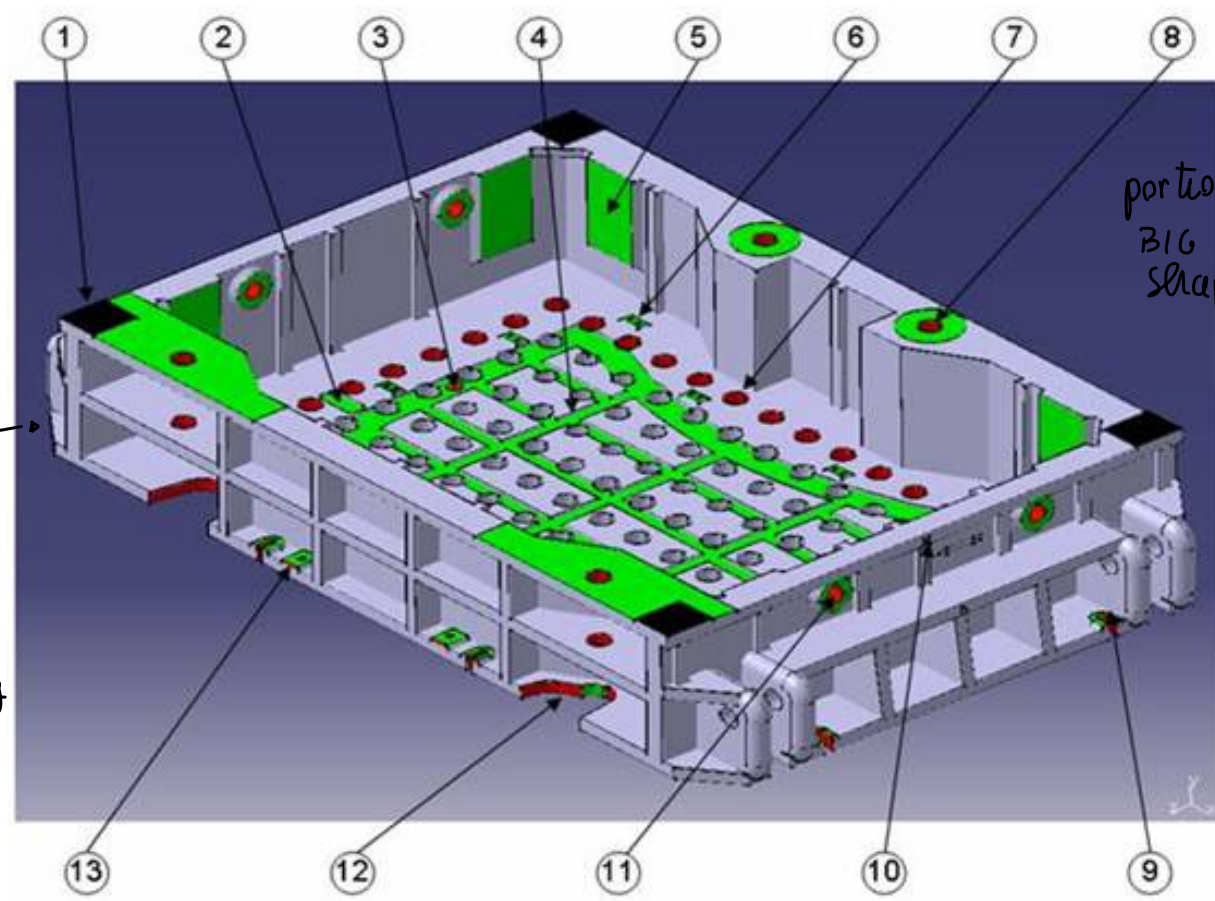
✓ PROCESS/ DECISION we need Rules to
FORMALIZE -- computer compatible PART description
(Feature based)

↑↑
Formalize KNOWLEDGE + Train people, increase
experience



Form Feature and Form Feature Recognition

1m MACHINING
↓
complex object
coming from
CASTING +
MACHINING



portion of a
BIG die for
shaping car
bolts

repeating
features --
IF you can
classify the
features and
define the machining
Rules → OBTAIN
AUTOMATIC
PROC. PLAN



CAPP SYSTEMS: DISCUSSION

AUTOMATIZE PROC. PLAN
↓

Critical Factors

you can work in a sector with STABLE PROCESS KIND, I can create a general plan, NOT expecting changes in technical. IF changing tech, NOT RETRIEVAL syst. & change I need to UPDATE the PLAN always! ⇒ I can TRY to formalize KNOWLEDGE

▶ Manufacturing System Components

▶ Production Volume

→ changing the amount of PARTS to do -- the process plan can completely change!
FROM FEW PARTS → SAND CAST / LOT PARTS → permanent mold!
↳ additive manuf

▶ Number of Part Families

(consider also how to GROUP similar product! (consider also the VOLUME))

depending on VOLUME I can completely change the solution for same part!



CAPP SYSTEMS: DISCUSSION

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(Represent our part mix)

decompose the area in 3 Region

Average Number
of Parts for each
Part Family

small families
but high volume

(2)

Variant
CAPP

NOT well defined line

lot Families

⇒ Semi-Generative

(3)

Generative
CAPP

(1) → NOTHING? What approach
if low volume and small
families ⇒ FEW PRODUCTION,
Rare Redesign → NOT JUSTIFY
Variant OR Generative!

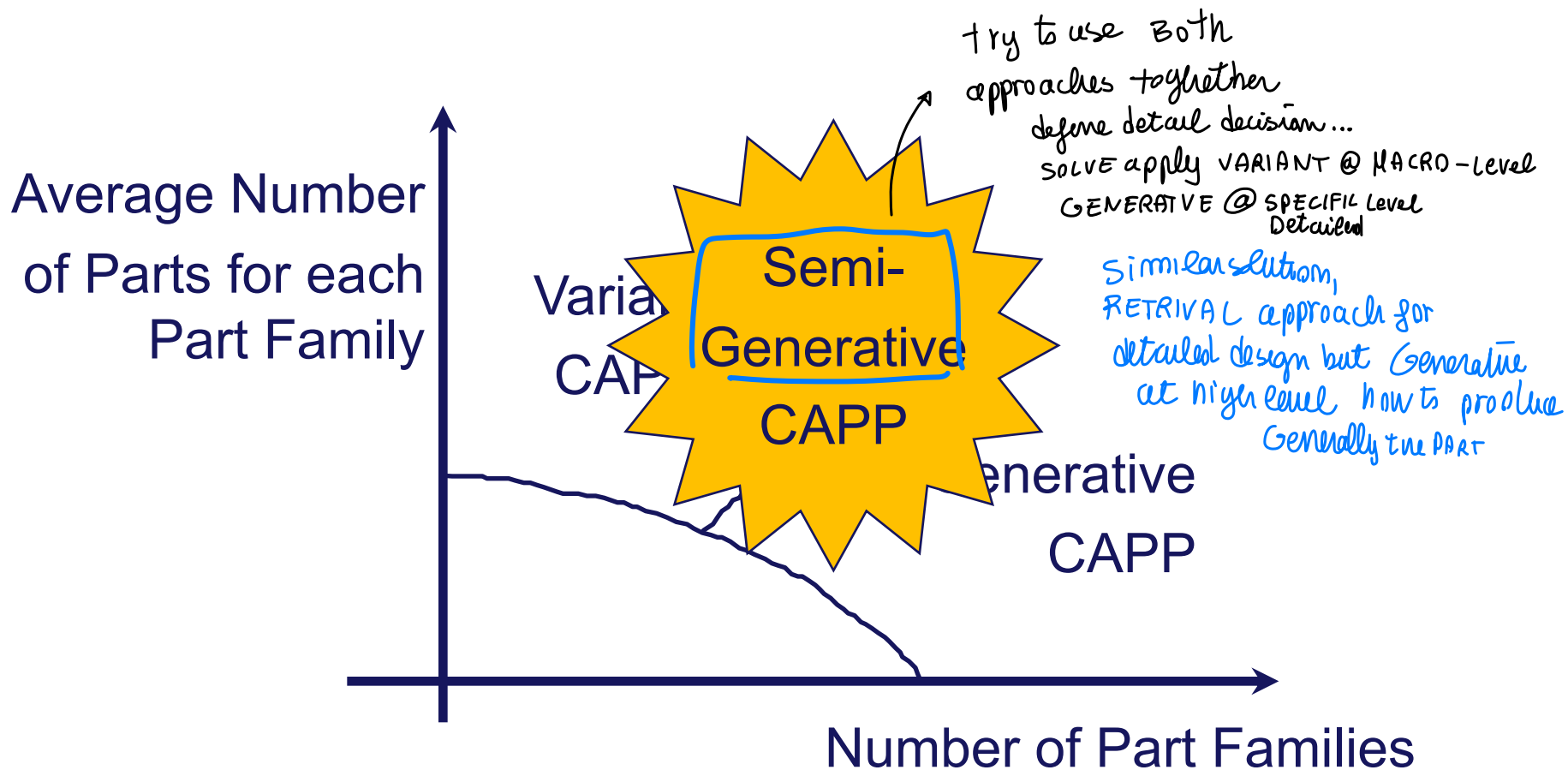
↓
DON'T JUSTIFY THE COST, TOO SMALL
PRODUCTION TO CHOOSE AUTOMATICALLY

(1)

Number of Part Families

(2) → VARIANT CAPP: probably company
WITH STANDARDIZE production → small component changes → very small # families,
standardize prod → usefull to have variant approach

(3) → small # part but many Families, opposite from (2), NOT standard production, each new part
IS A LOT DIFFERENT (NOT RETRIVAL system)





Semi-Generative CAPP

- ▶ Variant approach for process plan retrieval + Generative approach for standard process plan modification
- ▶ Generative approach for macro planning + Variant approach for detailed plan definition.



BENEFITS OF CAPP

the effort of CAPP...
HELP to take decisions

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- **Process standardization and optimization** – CAPP leads to more logical and consistent process plans than traditional process planning.
- **Better re-use of knowledge and improved training.**
- **Increased productivity of process planners**, by reduced lead time to prepare process plans.
- **Decreased time and cost for process plan maintenance.**
- CAPP programs can be interfaced with other application programs, such as cost estimating, work standards, and others.

Rules of Why making some decisions

IF system suggest decision, FASTER process planner!

"maintenance" in environment of MIX PART, each part has its PLUM PLAN, IF change technology I need to update ALL PART PLAN! (MAINTENANCE issue)

IF VARIANT: change inherent by all the PARTS
IF GENERATIVE: remove rule and recompute the plan

BEHIND... PART PROGRAMMING of CNC (CAM)
PROCESS PLANNING (CAPP)



There is Reasoning behind GEOMETRY of the PART

⇒ know something about Geom represent. of

SOLID/LINE/SURFACE on a computer



Geometrical computation

ISSUE to solve on SW creation